

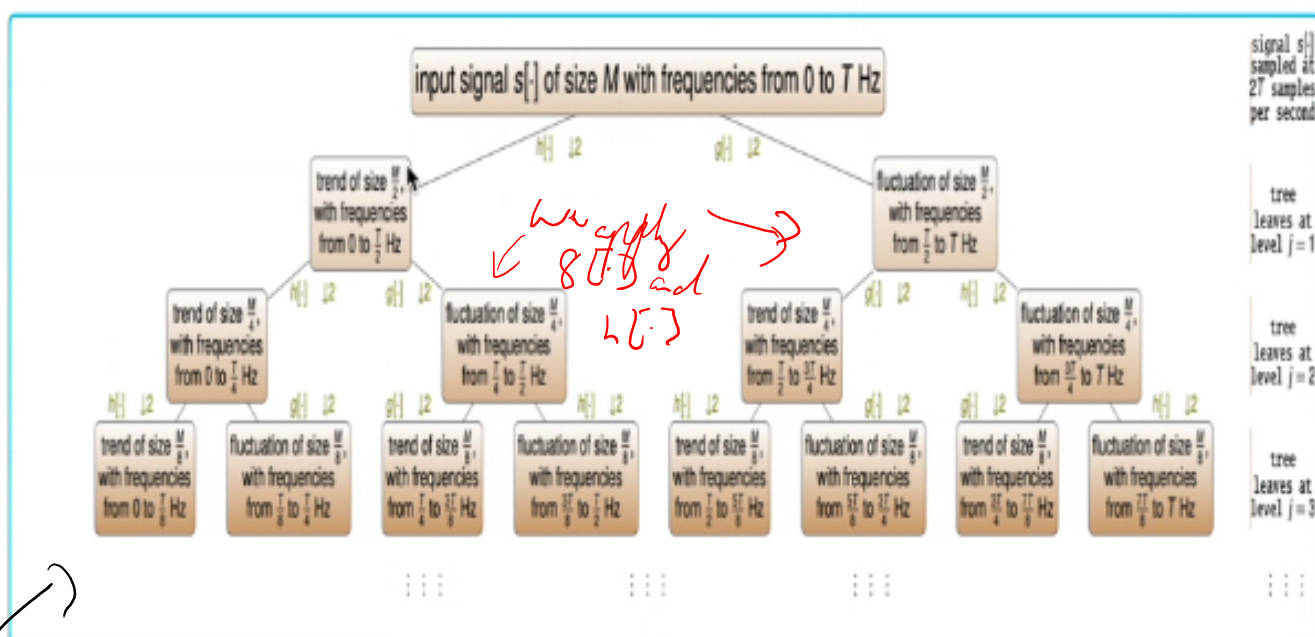
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26

DTWPT $\rightarrow$ Discrete-time wavelet-packet transform

$\hookrightarrow$ decompose both sides of the tree

--- Wavelet-Packets ---

Look at this DTWPT decomposition tree, exemplified for the level $j = 3$:



- Pay **MUCH** attention to the decomposition tree shown above. Notably, there is one curious detail regarding the application of the filters $h[\cdot]$ and $g[\cdot]$, as follows:

give a length resolution for all subband

when you have a band that's generated by a $g(t)$ filter $\rightarrow$ we previously do a downsampling, since its bandwidth has changed, but its max frequency hasn't. Due to that if we filter again

the result is a mirrored spectrum.
That's why we use $[E]$ on the
left and $[E^*]$ on the right, to avoid this
mirrored nature

This interaction between filters
is essential to maintain the natural
frequency ordering (NFO)

~~As the filter is used on the whole
signal~~

A level n of a DTWPT is
basically the same as level n in
the tree concatenated from left to right

DTWPT also conserve the energy and
the size of the original signal in
the DTWT

we wish pain before doing the
next level. We don't need to mix the
filter position for next algorithm

For the inverse, we need to invert the
the relevant algorithm and the apply the
reorder algorithm for the inverse. We do
the same as before, but the roots of
the relation on the parts we gave us

We could use DT & PT for the spectral notation

For time-frequency plan, we could get the margin and the frequency would range down

For the shift convention, we must first exchange the position for DT & PT and then calculate the shift and energy

Usually we would pad a signal with zeros to make it size for the next power of two. However, there are cases that padding could be much less, we could cut some parts of the signal to reach the closest power of two

The decomposition is done with the leave expanded not the whole DT & PT